Expt. 7 Transient states in the RLC series circuit

Ⅰ. Purpose

(1) To study the transient states in RC, RL and RLC circuits.

(2) To understand the physical meaning of the time constant; learn the method of measuring time constant.

Ⅱ. Apparatus

THMJ-1 AC physical experiment instrument, digital oscilloscope, connecting leads, etc..

Ⅲ. Principle

By combining the components R, L and C, we can form four different circuits of RC, RL, LC and RLC. When AC power is switched on or off, there is a change from one equilibrium state to another. During the change the circuit is in a transient state.

1. RC circuit

In the AC series circuit comprising a Resistor R and Capacitor C, the transient state is the charge/discharge process of the capacitor (see Figure 7 - 1). When the Switch K is turned to Position 1, the power source begins to charge the capacitor C. The loop equation and initial conditions are

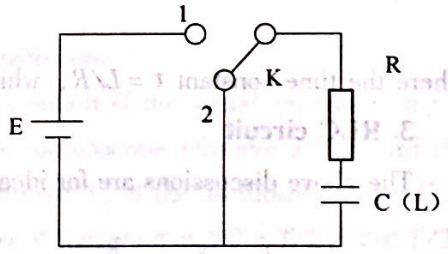


Figure 7 - 1 RC (RL) Circuit

$$\begin{cases} \dfrac{du_c}{dt} + \dfrac{1}{RC}u_c = \dfrac{E}{RC} \\ u_c(t)\big|_{t=0} = 0 \end{cases} \qquad (7-1)$$

Solving it, we find

$$u_c = E(1 - e^{-t/RC}) \qquad (7-2)$$

This expression shows that the voltage change across the capacitor is a curve increasing exponentially, shown as the solid line in Figure 7 - 2. The time for the voltage u_c to increases to $0.63E$, $\tau = RC$, is defined as the time constant of the circuit. This is an important physical quantity to characterize the speed of the transient state process.

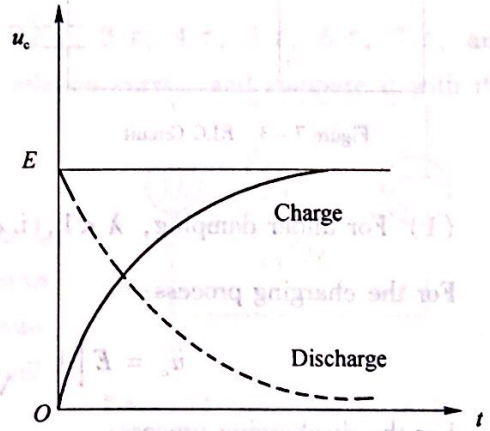


Figure 7 - 2 Charge and Discharge of a Capacitor

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When the Switch K is turned to Position 2, Capacitor C discharges through the resistor, with the loop equation and initial condition given by

$$\begin{cases} \dfrac{du_c}{dt} + \dfrac{1}{RC}u_c = 0 \\ u_c(t)\big|_{t=0} = E \end{cases} \tag{7-3}$$

Solve this to give

$$u_c = Ee^{-t/RC} \tag{7-4}$$

This expression shows that the voltage change across the capacitor is a curve decreasing exponentially, shown as the dashed line in Figure 7-2. The time constant τ can be determined as the time it needs for the curve to decrease to $0.37E$.

2. RL circuit

Replace C in Figure 7-1 by L, to produce an RL circuit. When Switch K is in Position 1, the current in the loop cannot switch instantaneously because of the self inductance effect of the inductor L, but it increases gradually to the maximum value E/R. In a similar way, when Switch K is in Position 2, current i gradually decreases from $i = E/R$ to 0. Similarly to the RC circuit analysis we can get.

For current increasing process:

$$i = \frac{E}{R}(1 - e^{-tR/L}) \tag{7-5}$$

For current decreasing process:

$$i = \frac{E}{R}e^{-t/\tau} \tag{7-6}$$

Where the time constant $\tau = L/R$, which determines the speed at which i increases.

3. RLC circuit

The above discussions are for ideal conditions. In reality, the capacitor and inductor themselves have an internal resistance and electrical energy is transformed into heat energy in these resistive components. The effect of the resistance is to introduce a damping term into the solution of the loop equation.

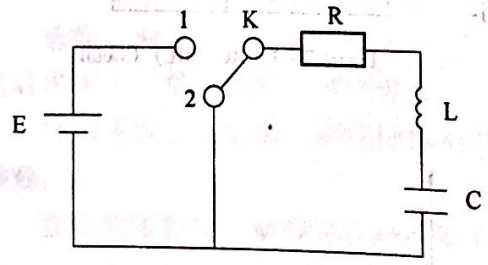


Figure 7-3 RLC Circuit

For the RLC circuit shown in Figure 7-3, we define the damping coefficient λ of the circuit as $\lambda = \frac{R}{2}\sqrt{\frac{C}{L}}$.

(1) For under damping, $\lambda < 1$, i. e. $R^2 < 4\frac{L}{C}$, we find

For the charging process:

$$u_c = E\left[1 - \sqrt{\frac{4L}{4L - R^2C}}\,e^{-t/\tau}\right]\cos(\omega t + \varphi) \tag{7-7}$$

For the discharging process:

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$$u_c = \sqrt{\frac{4L}{4L - R^2C}} E e^{-t/\tau} \cos(\omega t + \varphi) \qquad (7-8)$$

where, $\tau = \frac{2L}{R}$ is the time constant, $\omega = \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{R^2C}{4L}}$ is the angular frequency of the damped oscillation.

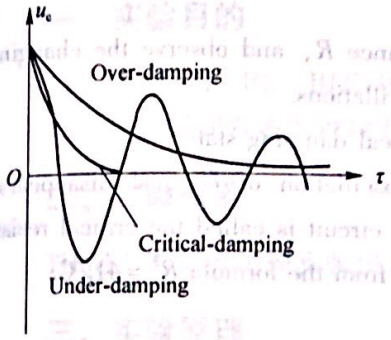


Figure 7 – 4 RLC Circuit Response to Step Voltage (Discharge Process)

u_c gradually decreases with time τ to zero as a damped oscillation, as shown by the curve in Figure 7 – 4. Such a state is called under-damped state.

(2) Critical damping state, i. e. $R^2 = 4L/C$, with the circuit resistance increasing to give a state of no oscillation. Here the damping process of u_c no longer has periodicity, as shown in the curve in Figure 7 – 4. The resistance here is called critical damping resistance.

(3) Over-damped state, i. e. $R^2 > 4\frac{L}{C}$, here the oscillation no longer appears, but the amplitude gradually decreases to zero. The variation in u_c with time is shown by the curve in Figure 7 – 4.

Ⅳ. Experimental

1. Transient state for an RC circuit

(1) Observe the change in voltage across the capacitor with time

Connect the circuit according to Figure 7 – 5, making the output of the signal generator an f = 500 Hz square wave. Connect it to the input terminal Y_1 of the oscilloscope. Observe and record the waveform of the square wave. Then connect U_C to the input terminal Y_2 of the oscilloscope. Use C as 0.47 μF. Change the resistance of R, to give the three values τ namely $\tau = RC = T/2$, $\tau \ll T/2$, $\tau \gg T/2$, where T is the input square wave period. Observe and record the waveform of U_C in the three conditions, and explain the variation of U_C.

(2) Measurement of the time constant τ

Change the resistance of R, separately making $T/2 = 2\tau$, 3τ, 4τ, 5τ, 6τ, 7τ, and measure the value of τ in each condition. Draw the $R - \tau$ relation curve, and compare it with the definition of τ, $\tau = RC$.

2. Transient state of the RL circuit

Replace the capacitor in Figure 7 – 5 with an inductor L of 10 mH, referring to steps in part 1, observe the waveforms of U_R and U_L for the three different values of τ. Vary the value of R over a suitable range, draw the $R - \tau$ relation curve, and compare it with the theoretical formula.

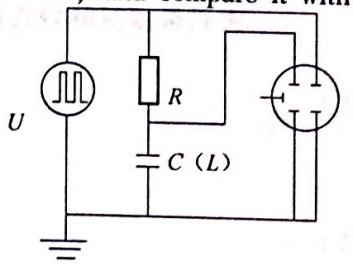


Figure 7 – 5 Connection Diagram for RC (RL) Circuit Transient States

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3. Transient state of the RLC circuit

(1) Connect your circuit according to Figure 7 – 6, taking the inductor L as 10 mH, capacitor C as 0.047 μF. Calculate the resistance ranges required for the three different damping states.

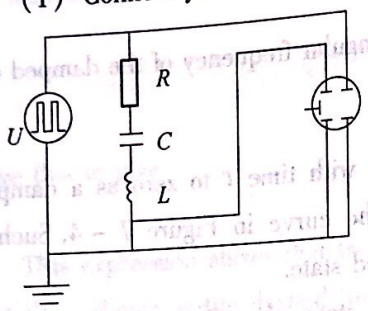


Figure 7 – 6 Connection Diagram for RLC Circuit Transient State

(2) Select an appropriate value of R, to produce an oscilloscope display of the total damped oscillation waveform.

1) Measure the period T and the attenuation time constant τ.

2) Change the resistance R, and observe the changing situation of the damped oscillations.

(3) Observe the critical damping state.

Gradually increase the value of R. When the waveform oscillation of U_C just disappears, i. e. the damping is in the critical state the total resistance of the circuit is called the critical resistance. Compare this resistance with the total resistance calculated from the formula $R^2 = 4L/C$.

(4) Observe the over-damping state:

Continue increasing the value of R, i. e. put the circuit into the over-damped state, and observe the influence of the various values of R on the waveform U_C.

V. Questions

1. In the RC circuit, when the square wave frequency is fixed the variation of resistance R leads to various waveforms. Why is this? In the case of the resistance R being fixed while the frequency is changed, could we get similar waveforms, why?

2. In the RLC circuit can we get damped oscillations at very high or very low frequencies of the square wave generator? If the frequency of the square wave generator is very high or low, then whether we can see the waveform the damped oscillation? Does the relationship between the period T and angular frequency ω change with change of frequency of the square wave?

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实验七　RLC串联电路的暂态过程

一、实验目的

(1) 研究RC、RL、RLC电路的暂态过程。

(2) 理解时间常数的物理意义，学会其测量方法。

二、实验仪器

THMJ-1型交流物理实验仪，数字式示波器，导线等。

三、实验原理

利用R、L、C元件进行组合，可以构成RC、RL、LC和RLC四种不同的电路。这些电路在接通或断开直流电源的瞬间，存在一个从一种平衡态转变到另一种平衡态的过程。这个过程即为暂态过程。

1. RC电路

在由电阻R及电容C组成的直流串联电路中，暂态过程即是电容器的充放电过程（见图7-1），当开关K打向位置1时，电源对电容器C开始一个充电过程，这时回路方程和初始条件可表示为

$$\begin{cases} \dfrac{du_c}{dt} + \dfrac{1}{RC}u_c = \dfrac{E}{RC} \\ u_c(t)\big|_{t=0} = 0 \end{cases} \tag{7-1}$$

可解得

$$u_c = E(1 - e^{-t/RC}) \tag{7-2}$$

该式表明电容器两端的充电电压是一条按指数规律增长的曲线，如图7-2的实线所示。当电压u_c上升到$0.63E$时，所对应的时间$\tau = RC$定义为电路的时间常数，是表征暂态过程进行快慢的一个重要物理量。

当开关K打向位置2时，电容器C通过电阻放电，回路方程和初始条件为

$$\begin{cases} \dfrac{du_c}{dt} + \dfrac{1}{RC}u_c = 0 \\ u_c(t)\big|_{t=0} = E \end{cases} \tag{7-3}$$

可解得

$$u_c = Ee^{-t/RC} \tag{7-4}$$

式(7-4)表明电容器两端的放电电压按指数规律衰减，如图7-2虚线所示。时间常数τ也可由此曲线衰减到$0.37E$所对应的时间来确定。

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2. RL 电路

把图 7－1 中的 C 换为电感 L 则构成 RL 电路。当开关 K 置于 1 时，由于电感 L 的自感作用，回路中的电流不能瞬间突变，只能是逐渐增大到最大值 E/R。同理，当开关 K 置于 2 时，电流 i 从 $i=E/R$ 逐渐衰减到 0。同 RC 电路的分析，可有

电流增加过程：

$$i = \frac{E}{R}(1 - e^{-tR/L}) \tag{7-5}$$

电流衰减过程：

$$i = \frac{E}{R}e^{-t/\tau} \tag{7-6}$$

其中，时间常数 $\tau = L/R$，它决定了 i 增长的快慢。

3. RLC 电路

以上讨论的都是理想化的情况，实际上，电容和电感本身都有电阻，而电阻性元件电能转化为热能，因电阻的主要作用就是把阻尼项引入到回路方程的解中。

对于图 7－3 所示的 RLC 电路，定义电路的阻尼系数 $\lambda = \frac{R}{2}\sqrt{\frac{C}{L}}$，则：

（1）阻尼较小时，$\lambda < 1$，即 $R^2 < 4\frac{L}{C}$，有

充电过程：

$$u_c = E\left[1 - \sqrt{\frac{4L}{4L - R^2C}}e^{-t/\tau}\right]\cos(\omega t + \varphi) \tag{7-7}$$

放电过程：

$$u_c = \sqrt{\frac{4L}{4L - R^2C}}Ee^{-t/\tau}\cos(\omega t + \varphi) \tag{7-8}$$

式中，$\tau = \frac{2L}{R}$ 为时间常数，$\omega = \frac{1}{\sqrt{LC}}\sqrt{1 - \frac{R^2C}{4L}}$ 为衰减振荡的角频率。

u_c 随时间 t 以衰减振荡的方式逐渐衰减至零，如图 7－4 中的曲线所示。这种过程也叫欠阻尼状态。

（2）临界阻尼状态，即 $R^2 = 4L/C$，回路电阻增大到刚刚不出现振荡时的状态。此时 u_c 的变化过程不再具有周期性，如图 7－4 中曲线所示。这时的电阻值称为临界阻尼电阻。

（3）过阻尼状态，即 $R^2 > 4\frac{L}{C}$，此时已不再出现振荡，而是缓慢衰减到零。u_c 随时间变化的规律如图 7－4 中曲线所示。

四、实验内容及数据处理

1. RC 电路的暂态过程

（1）观测电容器上电压随时间的变化关系

按图 7－5 连线，令信号发生器输出频率 $f = 500\text{Hz}$ 的方波，并接入示波器 Y_1 输入端，观察记录方波的波形，再将 U_C 接到示波器 Y_2 输入端，电容 C 取 0.47μF。改变 R 的阻值，

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使τ分别为$\tau = RC = T/2$，$\tau \ll T/2$，$\tau \gg T/2$，T是输入方波信号的周期，观察记录这三种情况下U_C的波形，并解释U_C的变化规律。

(2) 测量时间常数τ

改变R的阻值，分别使$T/2 = 2\tau$，3τ，4τ，5τ，6τ，7τ，利用示波器测量每种情况下的τ值，作出$R-\tau$关系曲线，并与τ的定义$\tau = RC$进行比较。

2. RL 电路的暂态过程

把图7-5中的电容用10mH的电感L代替，参照实验内容1中的步骤，观测三种不同τ值情况下，U_R和U_L的波形，并任意改变R值，作出$R-\tau$关系曲线，并与理论公式进行比较。

3. RLC 电路的暂态过程

(1) 按图7-6连线，取电感L为10mH，电容C为0.047μF，计算三种不同阻尼状态所对应的电阻值范围。

(2) 选择合适的R值，使示波器上出现完整的阻尼振荡波形。

1) 测量振荡周期T及衰减时间常数τ。

2) 改变R的阻值，观察振荡波形的变化情况。

(3) 观察临界阻尼状态：

逐渐加大R值，当U_C的波形刚刚不出现振荡时，即处于临界状态，此时回路的总电阻就是临界电阻。把该临界电阻与用公式$R^2 = 4L/C$计算出来的总阻值进行比较。

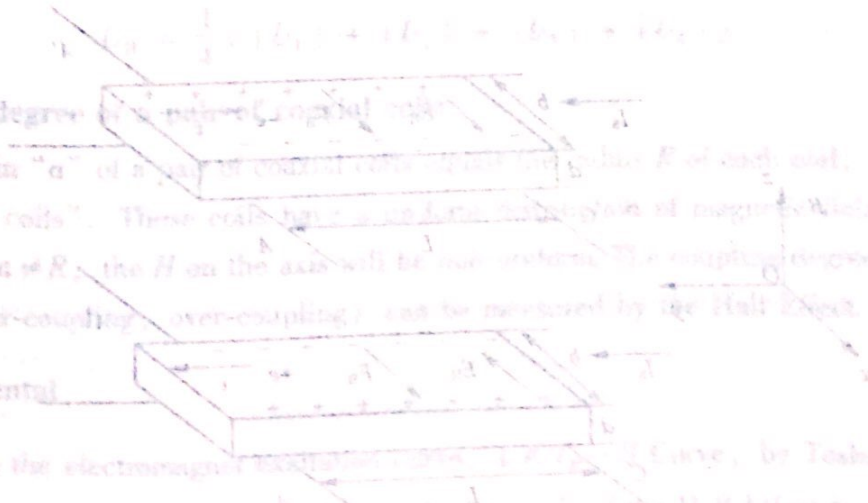
(4) 观察过阻尼状态：

继续加大R值，即电路处于过阻尼状态，观察不同R对U_C波形的影响。

五、思考题

1. 在RC电路中，当方波频率f一定而电阻R的数值改变时，为什么会有各种不同的波形？如果电阻R一定而方波频率f改变，那么会得到类似的波形吗？为什么？

2. 在RLC电路中，方波发生器的频率很高或很低，那么我们能观察到阻尼振荡的波形吗？振荡周期T与角频率ω的关系会因方波频率的变化而发生变化吗？



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实验十三　RLC 串联电路的暂态过程

1. RC 电路的暂态过程

已知：方波频率 $f=500\text{Hz}$，电容 $C=0.47\mu\text{F}$

①观察记录方波的波形，用坐标纸作图。

②改变 R 的阻值，使 τ 分别为 $\tau \ll T/2$；$\tau = T/2$；$\tau \gg T/2$，T 是输入方波信号的周期，观察记录这三种情况下 U_C 的波形，用坐标纸作图。

	$\tau=RC=T/2$	$\tau \ll T/2$	$\tau \gg T/2$
τ			
R			

③改变 R 的阻值，分别使 $T/2=3\tau, 4\tau, 5\tau, 6\tau, 7\tau$，测量每种情况下的 τ 值，用作图法讨论 τ 随 R 的变化规律，并与 τ 的定义 $\tau=RC$ 进行比较。

观察记录下列情况下的波形，用坐标纸作图。

	$T/2=3\tau$	$T/2=4\tau$	$T/2=5\tau$	$T/2=6\tau$	$T/2=7\tau$
R					
理论 τ 值					
实测 τ 值					

2. RL 电路的暂态过程

已知：方波频率 $f=500\text{Hz}$，

电感 $L=100\text{mH}$

参照实验内容 1 中的步骤，

观测三种不同 τ 值情况下，U_R 和 U_L 的波形，用坐标纸作图。

3. *RLC* 电路的暂态过程

已知：方波频率 $f=500\text{Hz}$，电感 $L=10\text{mH}$，电容 $C=0.047\mu\text{F}$

①利用公式计算三种不同阻尼状态所对应的电阻值范围。

	欠阻尼状态	临界阻尼状态	过阻尼状态
R 的范围（理论）			
R 的范围（实测）			

②选择合适的 R 值，使示波器上出现完整的阻尼振荡波形。测量振荡周期 T 及衰减时间。改变 R 的阻值，观察振荡波形的变化情况，并加以讨论。

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振荡周期 $T=$　　衰减时间 =

③逐渐加大 R 值，当 U_C 的波形刚刚不出现振荡时，即处于临界状态，此时回路的总电阻就是临界电阻。该临界电阻与用公式 $R^2=4L/C$ 所计算出来的总阻值进行比较。

④继续加大 R 值，即处于过阻尼状态，观察不同 R 对 U_C 波形的影响。

观察记录这三种状态下 U_C 的波形，用坐标纸作图。

思考题

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